BUKIDNON STATE UNIVERSITY
College of Technologies
Information Technology Department

Academic Early Warning System

Requirements Review Report

IT143 CO3 Major Assessment
Quality Assurance Project Management Framework

Document control	Details
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Review method	Static inspection of supplied assessment materials and local project artifacts
Document status	Review report prepared; stakeholder validation and baseline approval pending
Associated Trello task	Card 1 - Review AEWS Requirements

Executive summary

This report evaluates the clarity, completeness, consistency, traceability and verifiability of the requirements proposed for the Academic Early Warning System (AEWS). Seventeen requirement groups are defined across account management, academic-data processing, early intervention, reporting and software quality. Their verification criteria are mapped to the existing forty-case QA suite.

The review identifies seven issues requiring clarification or corrective action, particularly identity assurance, student-record access, prediction prerequisites and output semantics. The requirements are suitable as a working QA baseline following resolution and stakeholder review of the identified issues. This document does not certify implementation correctness or release readiness.

Personal names, actual review decisions and approval dates remain editable on the sign-off page. No stakeholder approval or execution result has been fabricated.

1. Review Purpose, Scope and Method

1.1 Purpose

The review establishes a structured basis for AEWS quality assurance before test execution. It identifies the system behavior to be verified, distinguishes observed implementation rules from proposed requirements, and records unresolved decisions that could otherwise produce ambiguous test results.

1.2 System context and scope

AEWS is a web-based academic monitoring and early-intervention application for Bukidnon State University. Instructors manage classes and academic records; AMU staff review referrals and intervention information; administrators manage institutional functions; students access their own portal and needs-assessment activities. The reviewed repository contains a React frontend, FastAPI services, MongoDB integration and prediction components [3-10].

The scope includes requirements for account access, ownership restrictions, academic imports, grade and attendance integrity, referrals, assessment forms, prediction, support routing, student profiles, administrative functions, reports, usability, performance and end-to-end consistency. Production deployment certification and claims of educational effectiveness are outside this review.

1.3 Materials and procedure

The activity guidelines and rubric establish the assessment context [1-2]. The project overview, routes, authorization helper, prediction logic and model documentation provide implementation evidence [3-10]. The existing QA pack provides the proposed case identifiers used for traceability [11]. No signed software requirements specification, stakeholder interview record or approved quality-target document was supplied for this review.

The review inventories features, formulates requirement statements, assigns verification criteria, checks source-level consistency, maps requirements to tests and records findings. No application-level execution, database inspection, model retraining or stakeholder acceptance session was performed.

1.4 Review criteria and interpretation

Criterion	Question applied during review
Clarity and consistency	Is the behavior understandable and compatible across workflows and sources?
Completeness and correctness	Are actors, inputs, valid states and exception cases defined and consistent with the intended project?
Verifiability	Can expected results be checked through an observable test or controlled inspection?
Traceability	Can the requirement be related to evidence, tests and a responsible QA activity?

Requirement IDs R01-R17 preserve the mapping used in the existing QA attachments. The word shall denotes proposed baseline behavior, not an assertion that the implementation already satisfies it. Priorities and acceptance targets in this report remain subject to project approval.

2. Requirements Register: Identity and Classes

Proposed requirements for stakeholder validation. Priority reflects QA impact; source references are listed in Section 11.

R01 Account registration and approval [High]

The system shall validate account registration data and apply the approved verification and approval lifecycle before granting the corresponding staff privileges.

Verification: Valid, invalid and duplicate registrations produce the defined outcomes. Approved, pending and rejected accounts receive access consistent with the approved state matrix.

Traceability: TC-01 to TC-03; TC-33. Evidence: [3-4, 8].

Review action: Confirm the authoritative account-state and rejection/reactivation rules.

R02 Authentication and session lifecycle [High]

The system shall require verified identity for protected operations and reject invalid or expired credentials and sessions. Password recovery shall follow a defined, single-use lifecycle.

Verification: Valid login succeeds; invalid credentials, expired/reset tokens and unauthorized requests fail. Post-logout access is tested against the documented invalidation policy.

Traceability: TC-04 to TC-05. Evidence: [4-5, 7].

Review action: Define server-side logout, token invalidation and recovery-token expiry; resolve finding F01.

R03 Role and record-level authorization [High]

The system shall authorize each protected read or modification using trusted identity, permitted role and record ownership or an explicitly approved scope.

Verification: An instructor cannot read or modify another instructor's protected class; students cannot read or change another student's private records; forbidden staff operations fail at the API.

Traceability: TC-06 to TC-08. Evidence: [4-8].

Review action: Approve the access matrix and address F01-F02; frontend navigation alone is insufficient.

R04 Class lifecycle management [High]

The system shall support authorized class creation, archiving and restoration while preserving the approved relationship between classes and enrollments.

Verification: Valid creation persists under the correct owner; invalid creation is rejected; archive/restore produces the agreed active/archived views without loss or unintended duplication.

Traceability: TC-09 to TC-10. Evidence: [3-4, 6].

Review action: Confirm effects on related enrollments, reports and referral visibility before accepting lifecycle results.

3. Requirements Register: Academic Data and Referral

Proposed requirements for stakeholder validation. Priority reflects QA impact; source references are listed in Section 11.

R05 Academic import and identity matching [High]

The system shall process supported classlist, gradesheet and attendance inputs into the intended class, student and term, and handle invalid inputs without silent data corruption.

Verification: Reconcile each synthetic source row with persisted/displayed values. Test wrong identities, duplicates, malformed/empty/unsupported files and applicable upload-size boundaries.

Traceability: TC-11 to TC-15. Evidence: [6].

Review action: Define supported templates, partial-failure behavior and duplicate/re-upload policy. Reviewed class-router limits are 10 MiB per file and 50 MiB per session.

R06 Grade integrity and term separation [High]

The system shall preserve whole grades, component values and previous-term records as distinct data and apply only approved academic calculations and rounding.

Verification: Independent expected values match each displayed term/component and derived value. Missing, invalid and boundary values are handled explicitly.

Traceability: TC-16 to TC-17. Evidence: [6, 9-10].

Review action: Approve grade scale, formulas, precision and missing-value conventions; address F04.

R07 Attendance calculation and completeness [High]

The system shall calculate attendance according to the approved attendance policy and distinguish a valid zero rate from absent source data.

Verification: Known present/session counts reconcile with the reported rate. Test 0%, 100%, missing and out-of-range values; record the denominator and treatment of each attendance status.

Traceability: TC-18. Evidence: [6, 9].

Review action: Confirm late/excused/blank-session handling and aggregation across formats.

R08 Automatic referral and AMU handoff [High]

The system shall create and display student referrals according to the approved referral rules, with correct student, class, reason and staff-scope linkage.

Verification: For the reviewed grade-based rule, whole midterm 2.50 through 5.00 is included; 2.49 is outside this trigger. Invalid 5.01 must not become a valid referral-grade input. Check repeat updates and component-only changes.

Traceability: TC-19 to TC-21. Evidence: [6-7].

Review action: The observed threshold requires policy confirmation. It does not define a predicted High Risk outcome.

4. Requirements Register: Assessment and Intervention

Proposed requirements for stakeholder validation. Priority reflects QA impact; source references are listed in Section 11.

R09 Needs-assessment configuration and submission [High]

The system shall present the applicable assessment version, validate required responses and persist answers for the correct student and invitation.

Verification: Complete answers save correctly; required omissions fail; legitimate false/no values remain valid. Invalid tokens and approved expiry/reuse cases are tested.

Traceability: TC-22 to TC-24; TC-34. Evidence: [7-8].

Review action: Confirm invitation expiry/reuse and behavior of existing invitations after form changes.

R10 Prediction prerequisites and provenance [High]

The system shall enforce the approved input prerequisites for each prediction entry point, preserve feature meaning and report interpretable outputs with identifiable model or fallback provenance.

Verification: AMU inference is not invoked when a required source is absent. Valid zero attendance and negative answers are retained. Fixed inputs/version produce consistent outputs; label meaning, probability scale and fallback source are checked.

Traceability: TC-25 to TC-28. Evidence: [7, 9-10].

Review action: Resolve differing sufficiency rules and distinct risk/intervention output categories in F03-F05.

R11 Intervention routing and communication [High]

The system shall persist authorized support-routing decisions and direct associated communications to the intended student using the configured delivery process.

Verification: Routing survives reload and is linked only to the intended referral. Controlled recipient tests verify content and delivery/queue failure reporting; accepted queuing is not described as confirmed receipt.

Traceability: TC-29 to TC-30. Evidence: [7].

Review action: Confirm routing authority, service options, retry policy and evidence of delivery.

R12 Student portal and profile management [High]

The system shall allow students to view their own permitted academic/intervention information and modify only approved profile fields after sufficient identity verification.

Verification: Expected own-student data is displayed; required profile information is enforced; allowed changes persist; invalid changes and identifier substitution do not expose or alter another student.

Traceability: TC-31 to TC-32; TC-08. Evidence: [4, 7].

Review action: Resolve student-route identity policy and restrict modification scope; see F02.

5. Requirements Register: Administration and Quality

Proposed requirements for stakeholder validation. Priority reflects QA impact; source references are listed in Section 11.

R13 Administration and auditability [High]

The system shall restrict administrative state changes to approved actors, preserve the account/form lifecycle and record audit events defined by the project.

Verification: Approval, rejection, archive/restore and form-change scenarios produce the intended state and audit information. Equivalent non-admin mutations are rejected.

Traceability: TC-33 to TC-34; TC-06. Evidence: [5, 8].

Review action: Confirm admin authority at every mutation endpoint and specify required event fields/retention.

R14 Dashboard and report reconciliation [High]

The system shall produce role-appropriate dashboards and exports consistent with authorized underlying records and selected filters.

Verification: Known synthetic totals/rows reconcile across views and exports. Empty results, long values, pagination and unauthorized-record exclusion are checked.

Traceability: TC-35 to TC-36. Evidence: [3-4, 6-8].

Review action: Agree counting units, aggregation rules, filter persistence and available export formats.

R15 Usability and responsive interaction [Medium]

The system shall support completion of the selected core tasks with understandable feedback and usable controls at the agreed viewport and input conditions.

Verification: Record an independent member's task walkthrough, keyboard checks, validation/loading/error states and narrow/desktop layouts. Critical actions remain reachable and readable.

Traceability: TC-37 to TC-38. Evidence: [4, 11].

Review action: Approve task-success criteria, target users and viewport/browser coverage; no usability pass is claimed.

R16 Measured response performance [Medium]

The system shall meet team-approved response-time targets under an explicitly recorded workload and environment.

Verification: Proposed baseline: twenty single-user runs with a 100-student synthetic class; report median/p95 and failures. Proposed limits: dashboard/search 3 s and single-student prediction 5 s, subject to approval.

Traceability: TC-39. Evidence: [11].

Review action: These are proposed targets, not measured results or assessment mandates; resolve F06.

R17 End-to-end consistency and regression [High]

The system shall preserve student identity and academic/intervention data through the core workflow and after relevant fixes.

Verification: Verify upload, referral, assessment, prediction, routing and reporting on one recorded build; repeat original failed steps and affected regression tests after fixes.

Traceability: TC-40 and linked defect cases. Evidence: [6-9, 11].

Review action: Track unresolved dependencies and failed fixes. Test execution completion does not constitute release approval.

6. Actor Permissions and Data Rules

6.1 Proposed authorization boundary

The following matrix expresses the intended validation boundary, not verified implementation behavior. Institutional scope and exceptions require approval. Each sensitive API must verify identity and authority independently of whether its page is visible in the interface.

Actor	Permitted scope to confirm	Negative case
Instructor	Own classes, enrollments, academic uploads, referral-related information and permitted reports	Another instructor's class or protected student data
AMU staff	Referrals and intervention records within the approved assignment/institutional scope	Unassigned or out-of-scope referral; admin-only changes
Administrator	Approved account, configuration and institutional reporting functions	Anonymous request or non-admin administrative mutation
Student	Own dashboard/profile and valid assessment invitation scope	Another student's ID; unrelated invitation or protected staff action

6.2 Academic and intervention rules

Student identity must remain stable across imports, enrollments, referrals and exports. A match requires the approved identity policy; a similar name alone is insufficient unless explicitly accepted and reviewed. Duplicate and partial imports require deterministic handling and clear feedback [6].

Whole midterm grade, component scores, final grade and prior-term values have different meanings. The reviewed grade-based referral boundary is inclusive from 2.50 to 5.00. Neither this rule nor an observed code comment substitutes for an approved institutional academic policy [6, 10].

Attendance 0% is a valid observation, whereas missing attendance is an unavailable source. False/no assessment answers can be legitimate completed responses; derived assessment scores alone do not establish completion. AMU prerequisite validation and the general model helper must implement the approved workflow contract [7, 9].

6.3 Data protection and evidence

Public Trello evidence must use synthetic or sanitized records. Do not attach real grades, assessment disclosures, credentials, connection strings or access tokens. Assessment screenshots and reports should retain only the context necessary to verify the test. This follows the activity's explicit prohibition on sensitive information in a public board [1].

7. Review Findings: Identity and Prediction

All findings are open. Severity is a proposed review priority, not an executed-test outcome or deployment certification.

F01 Trusted identity is not consistently enforced at the authorization helper [High]

Observation: The reviewed get_current_actor function accepts X-User-Id and X-User-Role when no usable bearer token is supplied. It checks presence and recognized role, but does not authenticate those header assertions within that branch [5].

Implication: A protected operation relying on this branch may treat client-declared identity as trusted. Source inspection establishes the branch; deployment exposure and effective controls have not been runtime-tested.

Recommended action: Specify and enforce trusted identity at every protected operation. Remove the fallback or ensure only an authenticated, validated gateway can supply it. Test forged headers, missing/expired tokens and cross-record access.

Affected coverage: R02-R03; TC-06 to TC-08; proposed supplemental authorization test. Proposed owner: Authentication/authorization developer; QA access tester.

F02 Student-record identity proof is undefined for public profile routes [High]

Observation: The reviewed public student profile read/update and dashboard functions locate records using a supplied student identifier. Their signatures do not show a per-request student authentication dependency. Student login validates a password, but the reviewed return does not itself establish a server-validated session for these subsequent calls [7].

Implication: Identifier knowledge must not be assumed to prove identity for private reads or updates. Effective deployment behavior needs controlled verification.

Recommended action: Define the student authentication/session or scoped-capability contract. Enforce ownership for private reads and changes. Execute cross-student and no-session tests and document any intentionally public fields.

Affected coverage: R03, R12; TC-08, TC-31 to TC-32. Proposed owner: Student portal/backend developer; QA access tester.

F03 Prediction sufficiency differs across entry points [High]

Observation: The AMU route requires midterm, attendance and genuine completed assessment data. The shared _can_make_prediction helper permits inference when at least two of four indicators are available [7, 9].

Implication: Different prerequisites may be intentional, but without an entry-point specification the same student may receive inconsistent availability decisions and ambiguous test expectations.

Recommended action: Approve a per-workflow prerequisite matrix. Distinguish valid zero/negative data from missing sources. Align each route/helper with that contract and test each missing-source combination.

Affected coverage: R07, R09-R10; TC-25 to TC-28. Proposed owner: AMU/prediction developer; requirements reviewer.

8. Review Findings: Data and Acceptance Criteria

All findings are open. Severity is a proposed review priority, not an executed-test outcome or deployment certification.

F04 Grade-domain validation needs a single authoritative rule [High]

Observation: The reviewed AMU helper treats finite non-negative midterm numbers as present without an upper bound in that check. The referral helper uses a specific 2.50-5.00 interval; the model documentation describes a 1.00-5.00 scale [6-7, 10].

Implication: A presence check is not a complete validity rule. Without consistent validation, zero, out-of-range or incorrectly scaled grades may reach downstream processing.

Recommended action: Approve the grade domain, missing-value sentinel and component conversion rules. Validate at import/update boundaries and before inference; test zero, negative, missing, non-finite and above-range values.

Affected coverage: R05-R06, R08-R10; TC-17, TC-19, TC-25. Proposed owner: Academic-data/prediction developer; requirements reviewer.

F05 Risk and intervention labels need explicit semantics [High]

Observation: The general model returns a risk result, while the AMU route also derives prediction_label values such as Academic Problem and External Factor and records a weight-score comparison basis [7, 9-10].

Implication: Risk category, intervention cause and probability refer to different concepts. A shared display label or undefined probability interpretation could mislead evaluation and users.

Recommended action: Document separate output fields and their meanings, label mapping, probability scale and model/fallback source. Reconcile interface wording and report logic. Use model evaluation evidence for accuracy claims.

Affected coverage: R10-R11, R14; TC-27 to TC-30, TC-35. Proposed owner: Prediction/UI developer; model evaluator.

F06 Non-functional acceptance targets lack approval evidence [Medium]

Observation: The supplied activity/rubric does not prescribe response-time or model-accuracy limits. The existing QA pack proposes performance targets, but no stakeholder-approved quality-target record was supplied [1-2, 11].

Implication: Usability, speed and prediction-quality outcomes cannot be objectively accepted without agreed metrics, conditions and scope.

Recommended action: Approve workload, environment, thresholds and measurement method before execution. Specify model-evaluation population, leakage controls, class counts and required metrics.

Affected coverage: R10, R15-R16; TC-27, TC-37 to TC-39. Proposed owner: QA coordinator; project stakeholders.

F07 Lifecycle and duplicate-data policies require consolidation [Medium]

Observation: Feature implementations exist for uploads, archive/restore, invitations and form configuration. A consolidated approved policy for duplicate imports, partial failures, invitation reuse/expiry and existing form versions was not supplied [6-8].

Implication: Undefined edge cases reduce reproducibility and can lead different testers to assign conflicting outcomes. This is a requirements-evidence gap, not proof that every implementation is defective.

Recommended action: Record authoritative decisions and owners. Update requirement criteria and fixtures, then rerun affected boundary and regression tests.

Affected coverage: R04-R05, R09, R13; TC-10, TC-13, TC-24, TC-34. Proposed owner: QA coordinator; feature owners.

9. Requirements Traceability and Verification

This matrix links the proposed requirement groups to the existing QA cases and the twelve-card Trello structure. A mapping demonstrates planned coverage; it does not demonstrate execution, a passing result or stakeholder acceptance [11].

Requirement	Existing test cases	Trello card(s)
R01 Accounts	TC-01 to TC-03; TC-33	4
R02 Sessions	TC-04 to TC-05	4
R03 Authorization	TC-06 to TC-08	4
R04 Classes	TC-09 to TC-10	5
R05 Imports	TC-11 to TC-15	5
R06 Grades	TC-16 to TC-17	6
R07 Attendance	TC-18	6
R08 Referral	TC-19 to TC-21	7
R09 Assessment	TC-22 to TC-24; TC-34	7; administrative checks in 4
R10 Prediction	TC-25 to TC-28	8
R11 Routing	TC-29 to TC-30	8
R12 Student portal	TC-31 to TC-32; TC-08	4
R13 Administration	TC-33 to TC-34; TC-06	4
R14 Reports	TC-35 to TC-36	9
R15 Usability	TC-37 to TC-38	9
R16 Performance	TC-39	9
R17 Integration	TC-40 and fix regressions	10

Supplemental coverage required by this review

Add distinct executions for untrusted identity headers (F01), student no-session/identifier substitution (F02), entry-point prerequisite combinations (F03), invalid-grade input paths (F04) and output/probability meaning (F05). Extend TC-06/08/17/25/27 with unique scenario identifiers or create additional cases. Each execution needs its own actual result and evidence.

Record build, environment, tester, date, inputs, expected result, actual result, status, evidence and linked defect. Card 11 records the eventual overall QA decision; it must not infer approval from the presence of this report.

10. Review Conclusion and Disposition

10.1 Conclusion

The review provides seventeen proposed requirement groups and planned coverage through forty existing QA cases. However, source-level identity concerns, differences in prediction prerequisites, incomplete grade-domain specification and undefined acceptance targets prevent an unconditional approval of the requirements baseline. Seven findings remain open. No implementation pass/fail counts or stakeholder approvals can be concluded from this static review.

Recommended disposition: revise and validate before baseline approval. This is the report's recommendation; the project group and authorized reviewers must record their actual decision below.

10.2 Conditions for baseline approval

Resolve the high-priority identity and data-contract findings; obtain approval of the role/ownership matrix and academic rules; define prediction inputs and outputs per workflow; approve measurable quality targets; update test cases and traceability; record the disposition of every remaining issue. Implementation fixes and passing regression results are evaluated in subsequent QA tasks.

10.3 Change control

Any accepted requirement change should record the affected requirement ID, reason, requester, decision maker, approval date and impacted test/card. Increase the baseline version after approval, retain the previous record and avoid overwriting historical execution evidence.

10.4 Review and sign-off

Project group / prepared by

Reviewer names and roles

Actual review date / reviewed build or source version

Decision: approve / approve with conditions / revise and resubmit

Agreed finding dispositions, conditions and follow-up owners

Authorized reviewer acknowledgement / actual approval date

A typed name here records the group's acknowledgement; this form does not imply a digital signature or instructor approval.

11. References and Evidence Register

[1] Bukidnon State University, College of Technologies, Information Technology Department. (n.d.). IT143 CO3 major assessment guidelines: Quality Assurance Project Management Framework using Trello. Supplied PDF, pp. 1-3.

[2] Bukidnon State University, College of Technologies. (n.d.). Rubric for group activity using Trello: Quality Assurance Project Management Framework/timeline creation. Supplied PDF, p. 1.

[3] AEWS project repository. (n.d.). README.md. Project overview and role-based feature inventory. Local artifact reviewed 7 November 2026.

[4] AEWS project repository. (n.d.). frontend/src/App.jsx. Application routes and staff ProtectedRoute role declarations. Local source reviewed 7 November 2026.

[5] AEWS project repository. (n.d.). backend/app/authz.py. create_access_token, decode_access_token, get_current_actor, require_roles and ensure_self_or_admin. Local source reviewed 7 November 2026.

[6] AEWS project repository. (n.d.). backend/app/routers/classes.py. Academic imports, identity validation, class lifecycle, _matches_midterm_referral_threshold and _build_auto_referral_reasons. Local source reviewed 7 November 2026.

[7] AEWS project repository. (n.d.). backend/app/routers/students.py and backend/app/routers/amu_staff.py. Student public routes, assessment, referrals, _missing_prediction_requirements, predict_referral and support routing. Registration of public routers checked in backend/app/main.py. Local source reviewed 7 November 2026.

[8] AEWS project repository. (n.d.). backend/app/routers/admin.py. Account, assessment configuration and institutional reporting routes. Local source reviewed 7 November 2026.

[9] AEWS project repository. (n.d.). backend/app/ai_model.py. _can_make_prediction, _select_prediction_profile, _predict_with_fallback and predict_student_risk. Local source reviewed 7 November 2026.

[10] AEWS project repository. (n.d.). backend/MODEL_DOCUMENTATION.md. Model profiles, input features, label mapping and evaluation workflow. Documentation reviewed 7 November 2026; statements require reconciliation with current code and approved policy.

[11] AEWS QA preparation artifacts. (2026). Complete assessment pack and Card 3 test-case workbook, TC-01 to TC-40. Project-prepared planning materials; not an independently approved specification or evidence of executed tests.

Evidence limitations

References identify local artifacts and named functions so review observations can be reproduced. Source contents may change after review. No commit identifier or signed requirements baseline was supplied; the actual reviewed build/version must be entered at sign-off. External standards compliance, institutional policy approval and live-system security were not assessed.